Supplementary Material

Why structural divergence varies among residues in enzyme evolution: contributions of mutation, stability, and activity constraints

Julian Echave & Mathilde Carpentier

S1 Toy model of mutational effects on structure, stability, and activation energy

This section derives the mutational effects on structure, stability, and activation energy for a minimal model: a single active-site residue connected to two regions of the protein scaffold (Figure~S1). We then show that the LFENM matrix formalism (Eqs.~4–6 of the main text) reproduces the results.

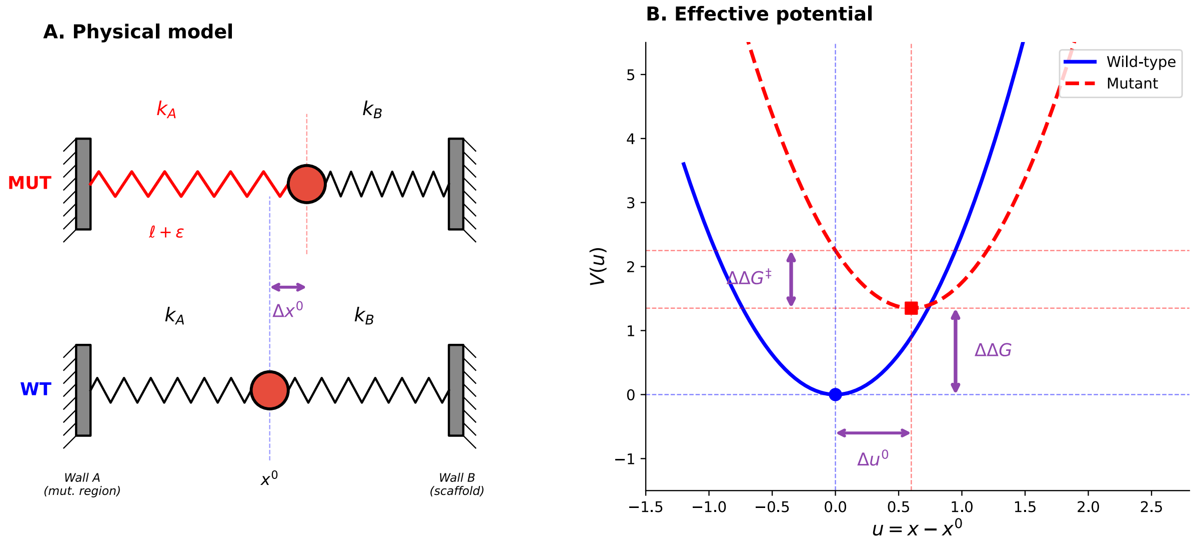

Figure S1: **Toy illustration of how LFENM calculates mutational effects on structure, stability, and activation energy.** (A) Physical model. A single residue (red circle) is connected by harmonic springs to the mutation region (Wall A, spring constant k_A) and the scaffold (Wall B, spring constant k_B). Bottom: wild-type equilibrium at x^0. Top: a mutation perturbs the natural length of the k_A spring by ε (red), shifting the equilibrium by Δx^0. (B) Effective potential energy as a function of displacement $u = x - x^0$. The wild-type potential (blue, solid) is minimised at $u = 0$; the mutant potential (red, dashed) is minimised at $u = \Delta u^0$. The vertical arrows indicate $\Delta\Delta G$ (stability cost) and $\Delta\Delta G^\ddagger$ (activation energy cost).

Setup

Consider a single bead (representing an active-site residue) at position x, connected by two harmonic springs to fixed walls representing rigid parts of the protein scaffold (Figure~S1A). Wall~A, at position A, represents the region of the protein containing the mutation site; Wall~B, at position B, represents the rest of the scaffold. The springs have force constants k_A and k_B and natural lengths $\ell_A = x^0 - A$ and $\ell_B = B - x^0$, where x^0 is the wild-type equilibrium position.

The wild-type potential energy is

$$V_{\mathrm{wt}} = \frac{k_A}{2}(x - A - \ell_A)^2 + \frac{k_B}{2}(B - x - \ell_B)^2. \tag{S1}$$

Defining the displacement from the wild-type equilibrium, $u = x - x^0$, we obtain:

$$V_{\mathrm{wt}} = \frac{k_A + k_B}{2}\, u^2. \tag{S2}$$

The wild-type minimum is at $u = 0$ with $V_{\mathrm{wt}}(0) = 0$.

Mutation

A mutation in region~A is modelled as a perturbation ε to the natural length of the k_A spring: $\ell_A \to \ell_A + \varepsilon$. The mutant potential is

$$V_{\mathrm{mut}} = \frac{k_A}{2}(u - \varepsilon)^2 + \frac{k_B}{2}\, u^2, \tag{S3}$$

which can be rewritten as

$$V_{\mathrm{mut}} = \frac{k_A + k_B}{2}\, u^2 - k_A\varepsilon\, u + \frac{k_A}{2}\, \varepsilon^2. \tag{S4}$$

The mutation adds a linear force $k_A\varepsilon$ pulling the bead toward a new position, plus a constant stress energy $k_A\varepsilon^2/2$. The quadratic term $(k_A + k_B)/2$ is unchanged: the mutation shifts the parabola horizontally and vertically but does not change its curvature (Figure~S1B). This is because the mutation perturbs the spring length, not the spring constant, so the Hessian is identical for wild type and mutant.

Structural change

Minimising V_{mut} with respect to u:

$$(k_A + k_B)\, u = k_A\, \varepsilon, \tag{S5}$$

the mutant equilibrium displacement is

$$\Delta u^0 = \frac{k_A\, \varepsilon}{k_A + k_B}. \tag{S6}$$

The bead shifts toward the position favoured by the mutated spring, but the opposing spring k_B prevents it from reaching $u = \varepsilon$. The compromise position Δu^0 lies between 0 and ε: closer to 0 when the coupling to the scaffold is strong ($k_B \gg k_A$) and closer to ε when the coupling to the mutation region is strong ($k_A \gg k_B$).

Stability change ($\Delta\Delta G$)

Because the Hessian is the same for wild type and mutant, the shape of the potential is unchanged by the mutation. The thermal fluctuations around each minimum are therefore identical, and the entropic contributions to the free energy cancel exactly. The free energy difference between mutant and wild type thus reduces to the difference in potential energy at their respective minima:

$$\Delta\Delta G = V_{\mathrm{mut}}(\Delta u^0) - V_{\mathrm{wt}}(0). \tag{S7}$$

Evaluating V_{mut} at Δu^0:

$$\Delta\Delta G = \frac{k_A}{2}(\Delta u^0 - \varepsilon)^2 + \frac{k_B}{2}(\Delta u^0)^2. \tag{S8}$$

Substituting Eq.~(S6):

$$\Delta\Delta G = \frac{k_A\, k_B}{2(k_A + k_B)}\, \varepsilon^2. \tag{S9}$$

This can be decomposed as

$$\Delta\Delta G = \underbrace{\frac{k_A}{2}\, \varepsilon^2}_{\text{stress}} - \underbrace{\frac{k_A^2}{2(k_A + k_B)}\, \varepsilon^2}_{\text{relaxation}}. \tag{S10}$$

The stress term $k_A\varepsilon^2/2$ is the energy cost if the bead were frozen at its wild-type position ($u = 0$). The relaxation term is the energy recovered by allowing the bead to shift to its new equilibrium. The stability cost $\Delta\Delta G$ is the residual stress that cannot be relaxed away: it arises from the conflict between the mutated and unmutated springs, which cannot be simultaneously satisfied.

Activation energy change ($\Delta\Delta G^\ddagger$)

The wild-type active-site geometry ($u = 0$) is assumed to be optimal for catalysis, meaning it coincides with the transition-state conformation. In the mutant, the active site has relaxed to $u = \Delta u^0$. To adopt the catalytically competent conformation, the mutant must be distorted from Δu^0 to $u = 0$. The cost of this conformational distortion is (Figure~S1B):

$$\Delta\Delta G^\ddagger = V_{\text{mut}}(0) - V_{\text{mut}}(\Delta u^0). \tag{S11}$$

Substituting Eq.~(S3) and Eq.~(S6):

$$\Delta\Delta G^\ddagger = \frac{k_A}{2}\,\varepsilon^2 - \frac{k_A\,k_B}{2(k_A+k_B)}\,\varepsilon^2 = \frac{k_A^2}{2(k_A+k_B)}\,\varepsilon^2. \tag{S12}$$

Connection to the LFENM matrix formalism

The derivations above used direct minimisation of the potential. We now show that the LFENM matrix formalism reproduces all three results. In the full LFENM, a mutation at residue j is encoded as a force vector $\mathbf{f}$, and the three mutational effects are calculated from the network matrix $\mathbf{K}$ (the Hessian of the wild-type energy) and its pseudo-inverse, the variance-covariance matrix $\mathbf{C}$:

$$\Delta\mathbf{r}^0 = \mathbf{C}\mathbf{f}, \tag{S13}$$

$$\Delta\Delta G = \frac{1}{2}\sum_{i\sim j} k_{ij}\,\delta l_{ij}^2 - \frac{1}{2}(\Delta\mathbf{r}^0)^T\mathbf{K}\,\Delta\mathbf{r}^0, \tag{S14}$$

$$\Delta\Delta G^\ddagger = \frac{1}{2}(\Delta\mathbf{r}_a^0)^T\mathbf{K}_{aa}^{\text{eff}}\,\Delta\mathbf{r}_a^0, \tag{S15}$$

where $\Delta\mathbf{r}_a^0$ is the active-site subset of $\Delta\mathbf{r}^0$ and $\mathbf{K}_{aa}^{\text{eff}} = \mathbf{C}_{aa}^{-1}$ is the inverse of the active-site block of $\mathbf{C}$.

For the one-bead model, $\mathbf{K}$, $\mathbf{C}$, and $\mathbf{f}$ reduce to scalars:

$$K = \frac{d^2V_{\text{wt}}}{du^2} = k_A + k_B, \qquad C = K^{-1} = \frac{1}{k_A+k_B}, \qquad f = k_A\,\varepsilon. \tag{S16}$$

Structural change. Eq.~(S13) gives:

$$\Delta u^0 = C\cdot f = \frac{k_A\,\varepsilon}{k_A+k_B}, \tag{S17}$$

confirming Eq.~(S6).

Stability change. Eq.~(S14) gives:

$$\Delta\Delta G = \frac{1}{2}k_A\varepsilon^2 - \frac{1}{2}(\Delta u^0)^2 K = \frac{k_A}{2}\,\varepsilon^2 - \frac{k_A^2\varepsilon^2}{2(k_A+k_B)}, \tag{S18}$$

confirming Eq.~(S10).

Activation energy change. In the toy model there is only one degree of freedom, the active-site coordinate u, so the active-site subspace is the entire space and no marginalisation over non-active-site coordinates is needed. Consequently $C_{aa} = C = 1/(k_A+k_B)$ and $K_{aa}^{\text{eff}} = C_{aa}^{-1} = k_A + k_B$. Eq.~(S15) with $\Delta\mathbf{r}_a^0 = \Delta u^0$ gives:

$$\Delta\Delta G^\ddagger = \frac{1}{2}(\Delta u^0)^2\,(k_A+k_B) = \frac{k_A^2\varepsilon^2}{2(k_A+k_B)}, \tag{S19}$$

confirming Eq.~(S12). In a multi-residue protein, marginalising over non-active-site coordinates generally yields $\mathbf{K}_{aa}^{\text{eff}} \neq \mathbf{K}_{aa}$, because $\mathbf{K}_{aa}^{\text{eff}}$ captures indirect couplings between active-site residues mediated through the rest of the protein; their equality here is a special feature of the one-bead model.

S2 Supplementary Tables

Table S1: Dataset entries and reference protein

M-CSA	PDB	Name	Source	Family
2	1btl_A	Beta-lactamase TEM	Escherichia coli	Class-A beta-lactamase family
15	1znb_A	Metallo-beta-lactamase type 2	Bacteroides fragilis	Metallo-beta-lactamase superfamily
71	1eug_A	Uracil-DNA glycosylase	Escherichia coli B	Uracil-DNA glycosylase (UDG) superfamily
98	1bsz_C	Peptide deformylase	Escherichia coli K-12	Polypeptide deformylase family
109	1d3g_A	Dihydroorotate dehydrogenase (quinone), mitochondrial	Homo sapiens	Dihydroorotate dehydrogenase family
148	1onr_A	Name not found	Escherichia coli	Transaldolase family
151	1q0n_A	2-amino-4-hydroxy-6-hydroxymethyldihydropteridine pyrophosphokinase	Escherichia coli	HPPK family
160	1ako_A	Exodeoxyribonuclease III	Escherichia coli K-12	DNA repair enzymes AP/ExoA family
164	1ruv_A	Ribonuclease pancreatic	Bos taurus	Pancreatic ribonuclease family
174	9pap_A	Papain	Carica papaya	Peptidase C1 family
216	1ca2_A	Carbonic anhydrase 2	Homo sapiens	Alpha-carbonic anhydrase family
252	1igs_A	Indole-3-glycerol phosphate synthase	Saccharolobus solfataricus	TrpC family
257	1xx2_A	Beta-lactamase	Enterobacter cloacae	Class-C beta-lactamase family
258	1sml_A	Metallo-beta-lactamase L1 type 3	Stenotrophomonas maltophilia	Metallo-beta-lactamase superfamily
290	1zio_A	Adenylate kinase	Geobacillus stearothermophilus	Adenylate kinase family
328	1lbm_A	N-(5&apos	Thermotoga maritima	TrpF family
351	1snz_A	Galactose mutarotase	Homo sapiens	Aldose epimerase family
362	1d6o_A	Peptidyl-prolyl cis-trans isomerase FKBP1A	Homo sapiens	FKBP-type PPIase family
376	1a4l_A	Adenosine deaminase	Mus musculus	Metallo-dependent hydrolases superfamily
394	1aj0_A	Dihydropteroate synthase	Escherichia coli	DHPS family
444	1cel_A	Exoglucanase 1	Trichoderma reesei	Glycosyl hydrolase 7 (cellulase C) family
462	1pnt_A	Low molecular weight phosphotyrosine protein phosphatase	Bos taurus	Low molecular weight phosphotyrosine protein phosphatase family
467	1cv2_A	Haloalkane dehalogenase	Sphingomonas paucimobilis	Haloalkane dehalogenase family
480	1czf_A	Endopolygalacturonase II	Aspergillus niger	Glycosyl hydrolase 28 family
597	1uch_A	Ubiquitin carboxyl-terminal hydrolase isozyme L3	Homo sapiens	Peptidase C12 family
681	1gq8_A	Pectinesterase	Daucus carota	Pectinesterase family
693	2rnf_A	Ribonuclease 4	Homo sapiens	Pancreatic ribonuclease family
749	1nml_A	No information found	Marinobacter nauticus	No information found
814	1glo_A	Cathepsin S	Homo sapiens	Peptidase C1 family
858	1mrq_A	Aldo-keto reductase family 1 member C1	Homo sapiens	Aldo/keto reductase family
877	1pbg_A	6-phospho-beta-galactosidase	Lactococcus lactis	Glycosyl hydrolase 1 family
908	1rtu_A	Ribonuclease U2	Ustilago sphaerogena	Ribonuclease U2 family
923	2acy_A	Acylphosphatase-1	Bos taurus	Acylphosphatase family
931	2pth_A	Peptidyl-tRNA hydrolase	Escherichia coli K-12	PTH family

Table S2: Dataset properties

M-CSA	N	<Id%>	<RMSD>	Length	AS size	AS RMSD	CATH	EC
2	6	43	1.04	263	6	0.35	alpha/beta	Hydrolases
15	6	35	1.16	225	8	0.40	alpha/beta	Hydrolases
71	8	43	1.08	223	4	0.53	alpha/beta	Hydrolases
98	10	37	2.15	168	7	0.44	alpha/beta	Hydrolases
109	4	44	1.31	359	7	1.78	alpha/beta	Oxidoreductases
148	5	53	1.06	316	5	0.57	alpha/beta	Transferases
151	4	47	1.14	158	4	1.08	alpha/beta	Transferases
160	10	34	1.49	256	7	0.38	alpha/beta	Hydrolases
164	5	48	1.09	123	5	0.50	alpha/beta	Lyases
174	12	43	1.01	212	4	0.75	alpha/beta	Hydrolases
216	5	39	1.09	256	6	0.30	alpha/beta	Lyases
252	6	35	1.75	245	7	0.61	alpha/beta	Lyases
257	10	45	0.94	359	6	0.43	alpha/beta	Hydrolases
258	4	38	1.97	263	7	0.87	alpha/beta	Hydrolases
290	7	47	1.10	217	5	0.55	alpha/beta	Transferases
328	4	33	1.47	193	2	0.64	alpha/beta	Isomerases
351	4	33	1.36	341	3	0.47	all-beta	Isomerases
362	7	46	0.92	107	6	0.43	alpha/beta	Isomerases
376	4	44	1.30	349	6	0.37	alpha/beta	Hydrolases
394	10	40	2.02	280	2	0.59	alpha/beta	Transferases
444	6	46	1.19	431	4	0.44	all-beta	Hydrolases
462	6	37	1.15	155	6	0.34	alpha/beta	Hydrolases
467	5	45	1.03	293	5	0.45	alpha/beta	Hydrolases
480	4	44	1.05	335	6	0.42	all-beta	Hydrolases
597	4	38	1.76	204	4	0.90	alpha/beta	Hydrolases
681	4	31	2.88	300	5	0.80	all-beta	Hydrolases
693	7	38	1.62	120	3	0.49	alpha/beta	Hydrolases
749	6	56	1.96	316	1	5.17	all-alpha	NA
814	4	46	0.72	215	4	0.37	alpha/beta	Hydrolases
858	16	38	1.88	322	4	0.68	alpha/beta	Oxidoreductases
877	23	37	1.52	447	2	0.48	alpha/beta	Hydrolases
908	5	45	1.45	107	4	0.49	alpha/beta	Lyases
923	4	34	0.99	98	2	0.49	alpha/beta	Hydrolases
931	7	39	1.40	193	4	0.43	alpha/beta	Hydrolases

N: number of family members; <Id%>: average family seq. identity; <RMSD>: average RMSD over all residues; Length: number of residues of reference protein; AS size: number of active-site residues; AS RMSD: average RMSD over active-site residues; CATH: CATH class; EC: EC class

Table S3: MSA model parameters and SHAP decomposition

M-CSA	PDB	a_S	a_A	D^2	R	sd(ϕ_{mut})	sd(ϕ_{stab})	sd(ϕ_{act})
2	1btl_A	0.20	87.50	0.49	0.67	0.23	0.05	0.28
15	1znb_A	0.47	42.43	0.65	0.78	0.34	0.13	0.24
71	1eug_A	0.91	2.56	0.75	0.83	0.39	0.24	0.02
98	1bsz_C	4.03	33.30	0.81	0.82	0.36	0.41	0.04
109	1d3g_A	0.82	0.73	0.44	0.62	0.26	0.25	0.01
148	1onr_A	0.69	3.35	0.47	0.66	0.29	0.20	0.02
151	1q0n_A	1.22	0.79	0.62	0.73	0.28	0.27	0.01
160	1ako_A	0.45	103.52	0.61	0.73	0.25	0.15	0.23
164	1ruv_A	1.11	340.36	0.70	0.71	0.32	0.19	0.30
174	9pap_A	0.13	15.44	0.28	0.52	0.23	0.04	0.10
216	1ca2_A	1.30	16.69	0.51	0.64	0.22	0.27	0.08
252	1igs_A	0.33	82.73	0.53	0.69	0.22	0.10	0.25
257	1xx2_A	0.43	8.82	0.56	0.73	0.26	0.14	0.06
258	1sml_A	1.81	1.22	0.71	0.76	0.50	0.25	0.01
290	1zio_A	1.49	1.30	0.62	0.71	0.30	0.24	0.01
328	1lbm_A	0.71	16.41	0.35	0.54	0.19	0.17	0.05
351	1snz_A	0.65	183.94	0.46	0.61	0.23	0.19	0.18
362	1d6o_A	0.32	80.27	0.69	0.76	0.27	0.07	0.38
376	1a4l_A	0.09	31.44	0.20	0.43	0.23	0.03	0.16
394	1aj0_A	1.68	28.26	0.66	0.74	0.24	0.35	0.07
444	1cel_A	0.70	7.93	0.33	0.52	0.21	0.19	0.03
462	1pnt_A	0.65	22.77	0.58	0.73	0.27	0.19	0.13
467	1cv2_A	0.84	1.91	0.41	0.58	0.21	0.24	0.03
480	1czf_A	0.72	16.61	0.40	0.57	0.19	0.17	0.07
597	1uch_A	0.83	1.56	0.46	0.63	0.25	0.23	0.01
681	1gq8_A	1.42	4.76	0.54	0.65	0.26	0.36	0.02
693	2rnf_A	0.67	374.66	0.68	0.73	0.25	0.10	0.31
749	1nml_A	0.96	1.37	0.29	0.48	0.31	0.20	0.01
814	1glo_A	0.11	41.77	0.45	0.66	0.24	0.04	0.19
858	1mrq_A	1.86	2.26	0.61	0.69	0.27	0.41	0.01
877	1pbg_A	0.80	18.97	0.46	0.63	0.25	0.21	0.03
908	1rtu_A	0.30	981.17	0.69	0.67	0.27	0.07	0.46
923	2acy_A	1.21	43.56	0.42	0.57	0.21	0.21	0.11
931	2pth_A	0.38	188.97	0.58	0.71	0.27	0.12	0.31

a_S: stability selection parameter; a_A: activity selection parameter; D^2: deviance explained by MSA model; R: Pearson correlation between observed and MSA-predicted lRMSD; sd(ϕ_{mut}), sd(ϕ_{stab}), sd(ϕ_{act}): standard deviations of mutation, stability, and activity SHAP components across residues